A bulge+disk lens model for DESI238

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Abstract

We model the strong lens DESI238 (HST WFC3/IR, F140W, 80×80 pixels at $0.065''$) with GIGA-Lens' MAP→SVI→HMC sequence. The adopted model has an EPL+external-shear deflector, a two-component (bulge+disk) deflector light profile, and a two-component Sérsic source: 36 free parameters. It gives $\chi^2_\nu = 0.989$ and is preferred over the single-component deflector light by $\Delta\mathrm{BIC} \approx -1762$. The Einstein radius is $\theta_\mathrm{E} = 1.480 \pm 0.003''$ ($\hat{R} = 1.001$, ESS = 94,186). A single-source, single-lens-light fit of the same data gave $1.516 \pm 0.002''$; the $0.036''$ difference is a model systematic an order of magnitude larger than either statistical error, and it, not the quoted $\pm 0.003''$, is the number that should propagate downstream.

1 Data and model

The cutout is 80×80 pixels at $0.065''\,\mathrm{pix}^{-1}$, extracted from the `final_94_drz.fits` drizzle product in units of $\mathrm{e}^-\,\mathrm{s}^{-1}$, with $t_\mathrm{exp} = 1197.699\,\mathrm{s}$. The PSF is a 27×27 kernel at the same pixel scale, normalised to unit sum. Rendering uses $2\times$ supersampling.

The noise model is the GIGA-Lens `ForwardProbModel`, whose per-pixel dispersion

$$\sigma = \sqrt{\sigma_\mathrm{bkg}^2 + \max(m, 0)/t_\mathrm{exp}} \tag{1}$$

depends on the model image m and so is re-evaluated every step. We take $\sigma_\mathrm{bkg} = 0.0075\,\mathrm{e}^-\,\mathrm{s}^{-1}$, the median of six widely separated blank patches in the parent frame; the patch-to-patch range is 0.0070–0.0092, and §6 shows what that range does to χ^2_ν.

The model has three blocks:

- **Mass**: an elliptical power law (EPL) plus external shear, eight parameters.
- **Deflector light**: two Sérsic profiles, given *different* priors because a bulge and a disk are physically distinct and identical priors would add a label-switching mode for nothing.
- **Source light**: two Sérsic profiles with the same (wide) prior.

Inference is 1024-start Adam (350 steps), then a full-covariance Gaussian SVI ansatz (1500 steps, warmup-then-decay schedule peaking at 2×10^{-3}), then 50 chains of preconditioned HMC, 3000 draws after 1000 burn-in. Everything internal is in unconstrained space; the tables below are after applying the prior's bijector.

*Claude Code (Anthropic). An AI wrote the modelling pipeline in `claude/DESI238/model/`, ran the fits, and drafted this report. The pipeline is a driver for GIGA-Lens (giga-lens/gigalens), which does the rendering and inference; the vendored package was not modified. The HST observations, the choice of target, and the direction of the analysis at every branch point came from the repository owner. Every number traces to `runs/bulgedisk_long/` and to `RESULTS_v3.md`; the tables were transcribed from `params.csv` by hand and have not been checked programmatically against it.

2 The fit

The residual panel of Figure 1 is featureless at the level of the arcs. Two things in it are worth flagging rather than passing over:

- A one-pixel-period vertical striping survives in the core at $\pm 5\sigma$. An earlier round attributed this to a saturating diverging colormap on a smooth dipole, but that argument was checked against the *model*, not this residual, and a one-pixel period is not what a colormap artifact on a smooth field looks like. It is unresolved.
- The `lens_light1` disk has $R_{\rm sersic} = 2.59''$, larger than the $2.6''$ half-width of the cutout. It is extrapolating over most of its own scale length, and should be re-checked against a larger crop before its parameters are read physically.

3 Parameters

All values are posterior medians with $\pm$ the standard deviation, from $3000 \times 50 = 150{,}000$ draws. Positions are in arcseconds relative to the cutout centre; I_e is in $\mathrm{e^-\,s^{-1}\,pix^{-1}}$ at the half-light radius. $\hat{R}$ is the Gelman–Rubin statistic across the 50 chains, and ESS the bulk effective sample size.

Table 1: Mass model: elliptical power law plus external shear. Every parameter has $\hat{R} \le 1.002$; the non-convergence discussed in §7 does not touch this block.

Profile	Parameter	Median	σ	$\hat{R}$	ESS
EPL	$\theta_{\rm E}$ ($''$)	1.479 641	0.002 826	1.001	94 186
	γ	1.792 080	0.021 054	1.000	61 723
	e_1	0.023 899	0.006 020	1.002	95 960
	e_2	0.016 562	0.005 270	1.001	89 243
	x_0 ($''$)	0.013 761	0.002 966	1.001	48 625
	y_0 ($''$)	−0.005 085	0.003 031	1.001	44 694
Shear	γ_1	0.142 112	0.004 309	1.001	91 224
	γ_2	−0.015 412	0.003 681	1.001	86 078

Two features of Table 1 deserve comment. First, the external shear is large: $|\gamma_{\rm ext} = 0.143|$, some 33σ from zero, which is well above what a typical field produces and should be checked against the galaxies visible in the parent drizzle frame. Second, the mass parameters are strongly correlated (Figure 2); in particular γ_2 and e_2 have $r = +0.947$, so the marginals $e_2 = 0.017 \pm 0.005$ and $\gamma_2 = -0.015 \pm 0.004$ are emphatically not two independent measurements. This is the familiar ellipticity–shear degeneracy of EPL+shear, and it is the reason to quote $\theta_{\rm E}$, which is nearly orthogonal to it, rather than the shape parameters.

4 The source

The two components are an extended host and a compact knot. Their half-light radii differ by only $3.2\times$ ($0.209''$ against $0.066''$); component 0 looks far larger in Figure 4 than that ratio suggests because the display is stretched and because component 1, at $n = 0.52$, is sub-exponential and has no wings — its surface brightness falls sixteen orders of magnitude by $r = 0.5''$, so there is nothing

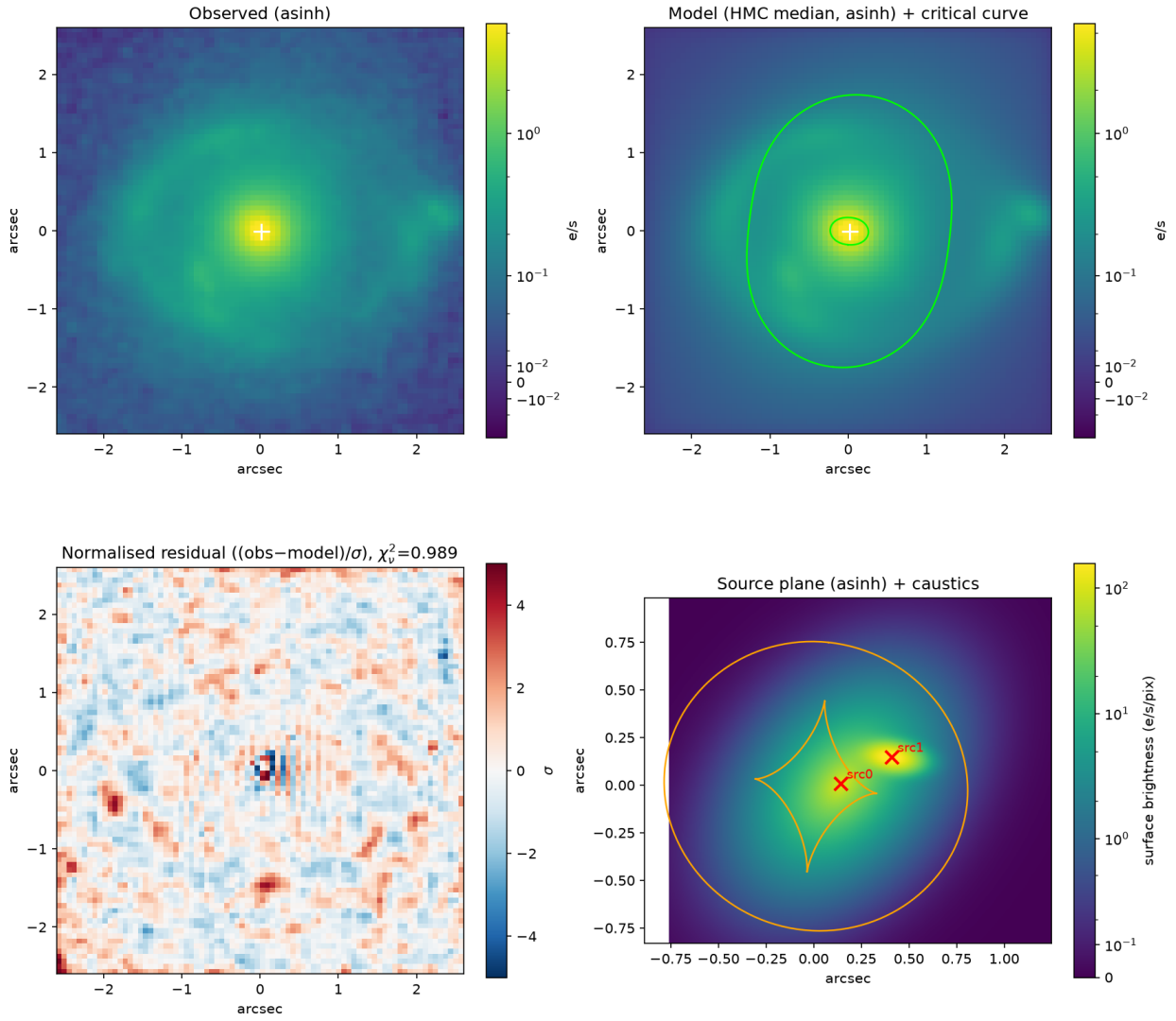


Figure 1: Observed image, model, normalised residual, and reconstructed source plane. The first, second, and fourth panels use an `asinh` stretch; the observed and model panels share a single normalisation so they are directly comparable by eye. The linear width of the stretch is $5\sigma_{\rm bkg}$, so the linear regime covers exactly those pixels consistent with background. Solid curves are the tangential and radial critical curves (panel 2) and their caustics (panel 4). The tangential critical curve has mean radius $1.545''$ against $\theta_{\rm E} = 1.480''$, a 4.4% difference — the expected offset for an elliptical, non-isothermal ($\gamma = 1.79$) deflector, and a check that the Jacobian orientation is right.

Table 2: Light profiles. The deflector's two components separate cleanly into a compact bulge and an extended disk, an 8.2× ratio in scale radius from priors that overlap at $\sim 1\sigma$. Rows in the source block marked † are discussed in §7.

Component	Parameter	Median	σ	$\hat{R}$	ESS
Deflector light					
bulge	I_e	265.609 558	21.464 956	1.012	3924
(`lens_light0`)	$R_{\rm sersic}$ ($''$)	0.316 435	0.018 986	1.012	4367
	$n_{\rm sersic}$	3.110 604	0.101 452	1.014	3687
	e_1	−0.006 533	0.003 503	1.019	3004
	e_2	−0.054 691	0.005 968	1.040	1331
	x_0 ($''$)	0.011 480	0.000 660	1.001	55 351
	y_0 ($''$)	0.008 185	0.000 582	1.000	75 225
disk	I_e	8.653 830	0.850 306	1.031	2017
(`lens_light1`)	$R_{\rm sersic}$ ($''$)	2.585 748	0.120 492	1.042	1594
	$n_{\rm sersic}$	2.091 850	0.143 315	1.096	602
	e_1	−0.083 190	0.005 385	1.004	10 766
	e_2	0.145 362	0.008 455	1.010	4888
	x_0 ($''$)	0.016 332	0.004 579	1.000	67 916
	y_0 ($''$)	0.003 584	0.004 166	1.001	61 822
Source light					
host	I_e	18.429 447	0.631 662	1.003	106 124
(`source0`)	$R_{\rm sersic}$ ($''$)	0.209 440	0.006 663	1.003	23 248
	$n_{\rm sersic}$	1.086 166	0.036 213	1.002	110 851
	e_1	−0.034 383	0.008 562	1.001	104 260
	e_2	0.186 871	0.007 973	1.001	105 610
	x_0 ($''$)	0.144 529	0.003 334	1.001	42 646
	y_0 ($''$)	0.005 654	0.002 808	1.002	24 599
knot	I_e	86.815 727	4.459 748	1.011	6745
(`source1`)	$R_{\rm sersic}$ ($''$)	0.065 510	0.002 623	1.002	16 883
	$n^{\dagger}_{\rm sersic}$	0.521 684	0.029 147	1.338	201
	e_1	0.318 078	0.023 257	1.009	5965
	e_2	−0.083 478	0.026 384	1.005	10 569
	x_0 ($''$)	0.411 215	0.010 253	1.001	61 416
	y_0 ($''$)	0.146 367	0.005 701	1.000	63 237

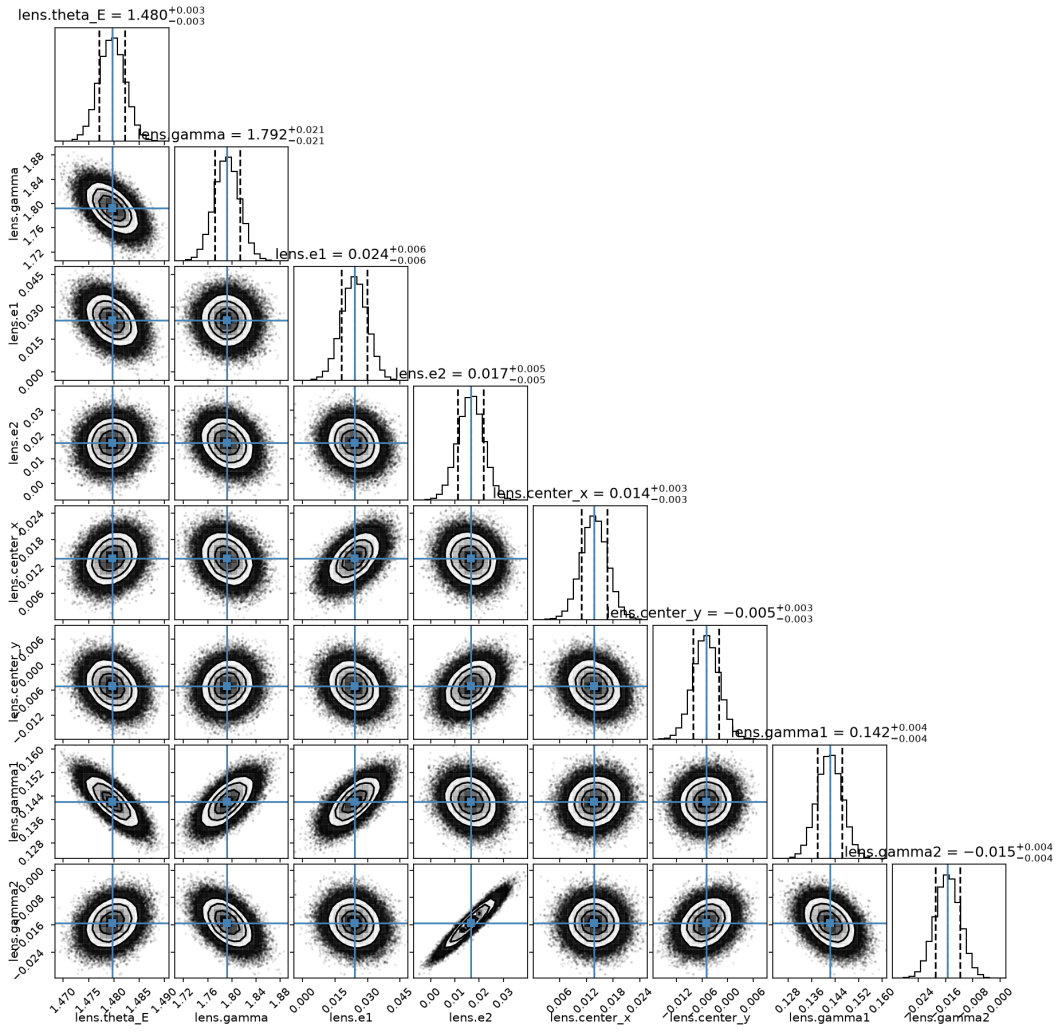


Figure 2: Posterior for the eight mass parameters. The dominant structure is the γ_2–e_2 ridge at $r = +0.947$; γ_1 correlates with $\theta_{\rm E}$ at $r = -0.776$ and with γ at $r = +0.661$. $\theta_{\rm E}$ itself is the tightest and least degenerate direction in the block.

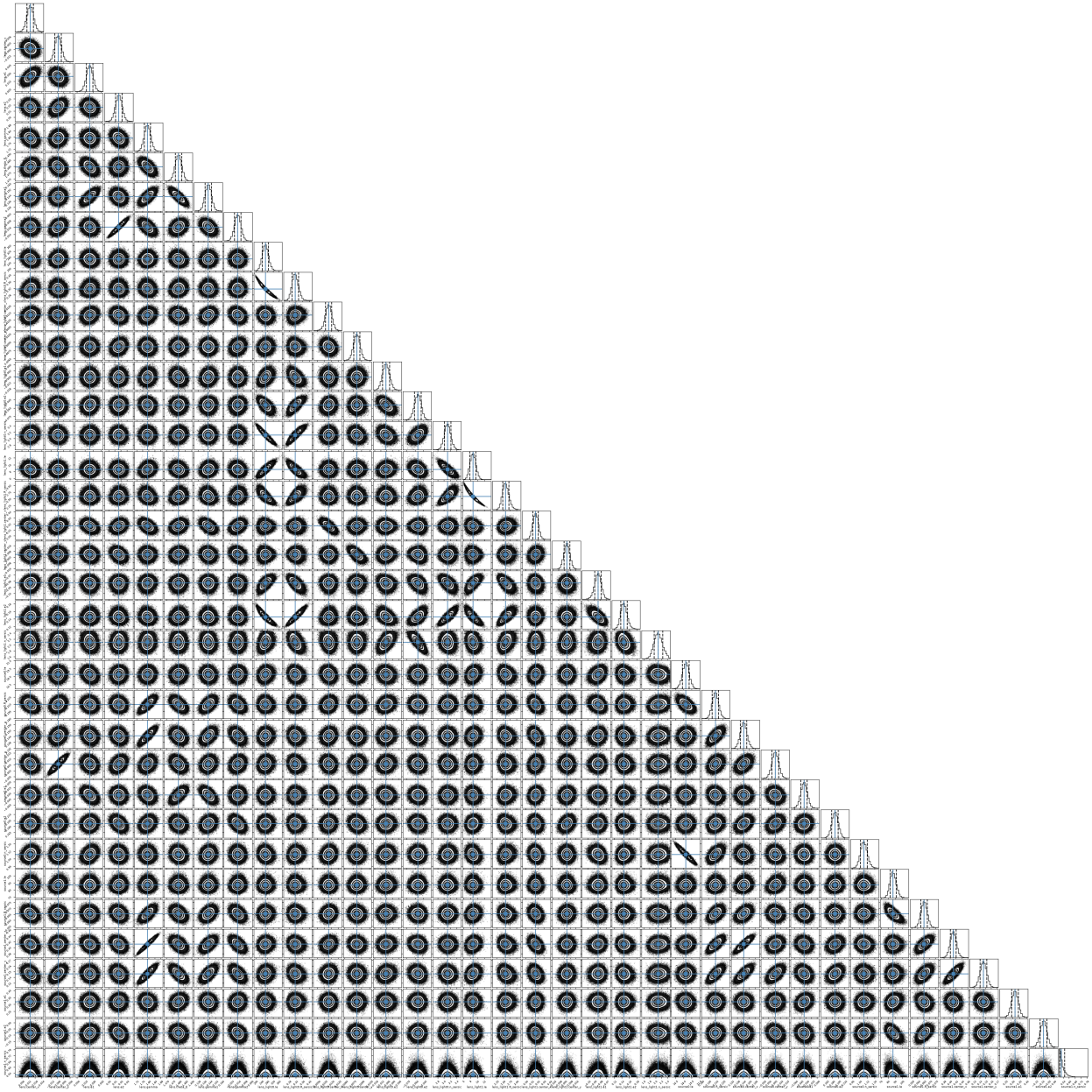


Figure 3: Full 36-parameter posterior. Reproduced at reduced size for completeness; the machine-readable summary is `runs/bulgedisk_long/params.csv`.

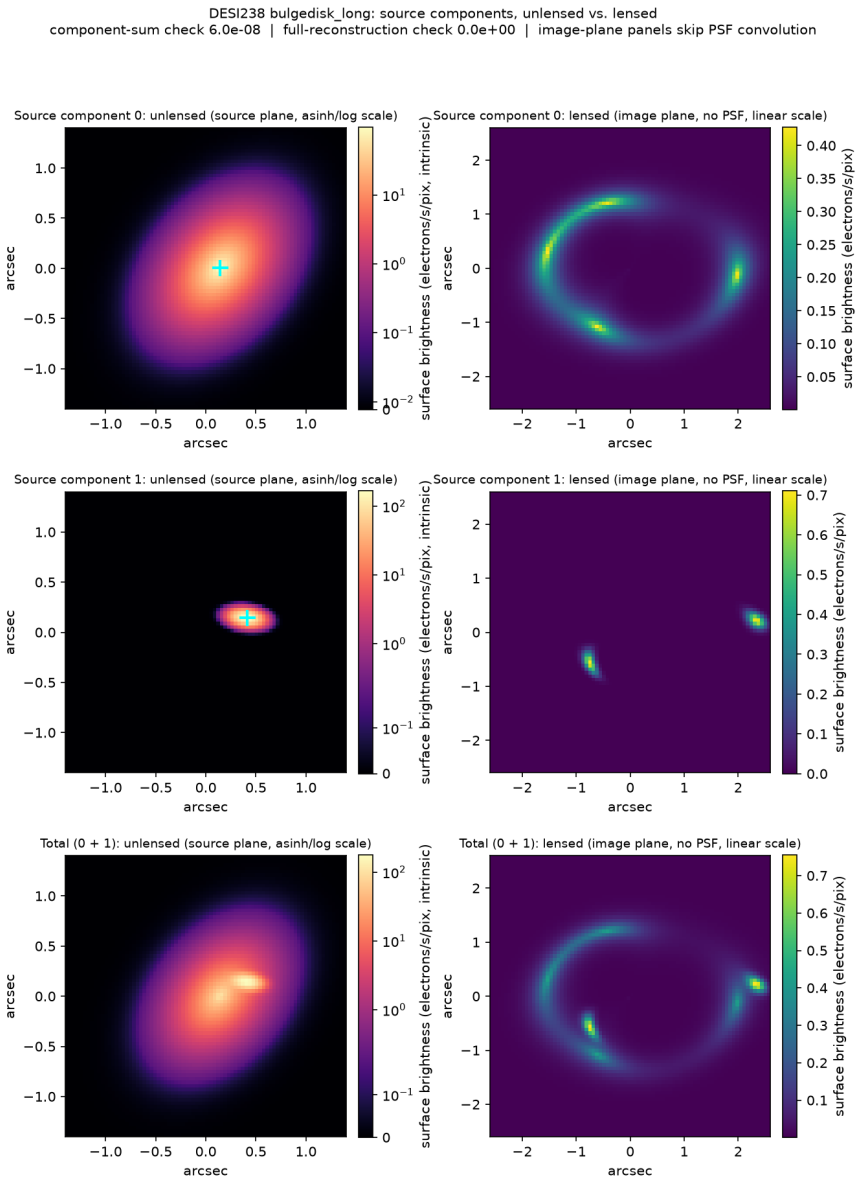


Figure 4: The two source components, unlensed (left column) and lensed (right). Component 0 lenses into the full ring; component 1 produces two detached counter-images. The sum of the two lensed components reproduces the full model image to $6 \times 10^{-8}\,\mathrm{e^-\,s^{-1}}$, confirming the decomposition is exact and not an approximation.

to render outside its core. Measured at fixed surface-brightness thresholds the ratio is 3.2, 4.0, and 4.8× at SB = 1.0, 0.1, and 0.01.

The knot is 4.2× brighter than the host in I_e and offset from it by 0.44″. That offset is what makes the second component earn its place: the astroid caustic in Figure 1 is only ∼0.5″ across, so the host, with $R_{\rm sersic} = 0.209''$, is comparable in size to the caustic and smears into a ring, while the knot straddles a fold and produces the detached images the data requires. A single source component can make one behaviour or the other, not both.

5 Model comparison

The bulge+disk deflector light was motivated by a coherent 1–3% radial S-curve in the core residual of the preceding fit — over-bright at $r < 1$ pix, too faint at 1.5–5.5 pix, too bright again at 6.5–11.5 pix. A single Sérsic is a one-shape family and cannot produce that pattern at any parameters. Table 3 gives the comparison against the otherwise identical 29-parameter model, evaluated at each run's point estimate over $N = 6400$ pixels.

Table 3: Model comparison. $\Delta \log L = +911.9$ for seven added parameters; ΔAIC $= -1810$, ΔBIC $= -1762$. BIC charges $\ln 6400 \approx 8.8$ per parameter against a return of ∼130 per parameter, so the penalty is irrelevant at this margin.

Model	k	χ^2	χ^2/N	$\chi^2/(N-k)$	$\log L$	BIC
1 lens light (`twosource_v2`)	29	8170.5	1.2766	1.2824	18 146.6	−36 039.0
2 lens light (`bulgedisk`)	36	6340.6	0.9907	0.9963	19 058.4	−37 801.4

Two qualifications. This is a maximum-likelihood comparison at point estimates, not a Bayes factor; a marginal likelihood would penalise the 36-parameter model's larger prior volume more heavily, though not by 1762. And because `ForwardProbModel`'s σ depends on the model image, the $-\sum \log \sigma$ normalisation is not constant between the two models. We checked its size rather than assuming it away: it contributes about −3 of the +912, so the comparison is carried by χ^2.

Adding the second deflector component took the core χ^2 ($r \le 8$ pix) from 7.09 to 2.97 and flattened the S-curve into sign-alternating scatter within $\pm 0.75\sigma$. The central pixel improved from -3.9σ to -1.5σ. Crucially, it did not do this by eating arc flux — the main risk, since a flexible light component near the Einstein radius can absorb arc surface brightness and bias $\theta_{\rm E}$. Aperture ratios on the four clumps went 1.09 → 1.03 (SW arc), 0.99 → 0.99 (E blob), 1.02 → 1.02 (W clump), and 0.91 → 0.83 (N arc), and $\theta_{\rm E}$ moved only 1.484 → 1.480″.

6 The noise scale

$\chi^2_\nu = 0.989$ should not be read as evidence that the noise model is now correct. $\sigma_{\rm bkg}$ is a single number drawn from six blank patches spanning 0.0070–0.0092, and χ^2_ν scales as $\sigma_{\rm bkg}^{-2}$:

Quote the improvement, not the absolute value. This makes building a per-pixel exposure map from the drizzle product's `WHT` extension *more* worth doing than before, not less: at $\chi^2_\nu = 1.3$ the noise model was a second-order worry, and at 0.99 it is the leading systematic. The frame has `NCOMBINE` = 3, so coverage genuinely varies across it and a scalar $t_{\rm exp}$ understates the noise wherever fewer exposures overlap.

Table 4: Sensitivity to the background dispersion. A ±10% error on $\sigma_{\rm bkg}$ is ±20% on χ^2_ν— larger than the distance from unity. The model comparison is untouched, because both models are evaluated under the same noise map and the scale factor largely cancels.

$\sigma_{\rm bkg}$	χ^2/N, 29 par	χ^2/N, 36 par	ΔBIC
0.0070	1.348	1.049	−1848
0.0075	1.277	0.991	−1762
0.0092	1.073	0.824	−1526

7 Convergence

Two distinct failures had to be separated here, and confusing them cost a full round of work.

The first was a diverged SVI stage. GIGA-Lens builds the HMC momentum distribution by inverting the SVI covariance in float32. When that covariance is badly conditioned the result is not an inverse: at the condition number of 8.8×10^{11} reached in an earlier run, $\max|C^{-1}C-I| = 55$ and the spectrum carried negative eigenvalues, so every chain sampled against an effectively indefinite mass matrix and *every* $\hat{R}$ failed at once. The covariance was still positive definite, so a PD test does not catch this — conditioning is the discriminant. The cause was an SVI learning-rate schedule that ramped upward, putting the largest steps last; replacing it with warmup-then-decay took the conditioning to 3.8×10^7 for the run reported here.

The second was ordinary under-mixing, and the two are easy to confuse because both raise every $\hat{R}$ at once. We separated them by brute force. The run reported here is identical to its predecessor in every respect — same seed, same priors, same MAP and SVI settings, verified by an identical SVI starting loss — except for chain length:

Table 5: Effect of chain length at fixed model and fixed preconditioner. $\hat{R}$ fell toward unity rather than plateauing near 3, which is the outcome that says the geometry is samplable and the chains were simply too short. No reparametrisation is needed.

	`bulgedisk`	`bulgedisk_long`
draws × burn-in	750 × 250	3000 × 1000
worst $\hat{R}$	3.242	**1.338**
parameters with $\hat{R} > 1.01$	33/36	**10/36**
χ^2_ν	0.991	0.989
SVI covariance condition	3.0×10^7	3.8×10^7

What still fails is confined and interpretable: the two deflector components' shape parameters, and `source1.n_sersic`. The latter is the worst in the run ($\hat{R} = 1.338$, ESS = 201) and is pinned at the 0.5 lower bound of its prior, sitting 0.4% into its range. A parameter on a boundary cannot mix, so its $\hat{R}$ is a symptom of the pinning and not independent evidence of anything. The floor itself is deliberate: GIGA-Lens uses $b_n = 1.9992n - 0.3271$, which is 3% wrong at $n = 0.5$ and 13.6% wrong at $n = 0.3$, so the fix is a different profile rather than a lower bound. This is the third consecutive round in which an $n_{\rm sersic}$ has pinned at 0.5, and it is the clearest unresolved defect in the model.

`lens_light1.n_sersic` sits at 80% of its Uniform(0.5, 2.5) prior and has the worst non-boundary $\hat{R}$ at 1.096. Widening it is a change to the prior's *support*, which silently rewrites the physical values

of stored runs (the checkpoints hold unconstrained samples), so it requires a prior-version bump rather than an edit in place. It was deliberately left out of the long run so as not to confound the one thing that run was measuring.

All eight mass parameters have $\hat{R} \leq 1.002$. The non-convergence never touched the lensing.

8 Result and outstanding work

$$\theta_{\rm E} = 1.480 \pm 0.003'', \qquad \hat{R} = 1.001, \quad \mathrm{ESS} = 94{,}186. \tag{2}$$

This is the first error bar from a multi-component fit of this system that survives its own diagnostic. It supersedes the $1.516 \pm 0.002''$ obtained from a single-source, single-lens-light model of the same data. The two do not agree within their quoted uncertainties, and they should not be expected to: the difference is a model change, not a sampling one, and the newer model is preferred by $\Delta\mathrm{BIC} \approx -1762$. **Treat the $0.036''$ gap as the honest systematic floor on $\theta_{\rm E}$ from light-profile choice.** It is an order of magnitude larger than either statistical error, and it is the number that matters for anything downstream.

Four things remain open, in the order in which they gate a stronger result:

1. **Whether the E blob is a counter-image.** This is the actual science question and it is still untested. A falsification run with $\theta_{\rm E}$ truncated below $\sim 1.7''$ would settle it; earlier attempts used a prior whose 99th percentile already admitted the four-image solution and so tested nothing. Deferring it was defensible while the source model was inadequate — a failed A/B could be blamed on the model — but at $\chi^2_\nu \approx 1$ that excuse is gone. Note that the E blob sits at $2.11''$, outside the tangential critical curve at $1.55''$, which bears directly on the interpretation.
2. **The per-pixel exposure map from `WHT`**, for the reasons in §6.
3. **The pinned $n_{\rm sersic} = 0.5$**, which needs a profile change, not a wider bound.
4. **The core striping**, which the earlier colormap explanation does not obviously cover.

Reproducing this

```
export PYTHONPATH=gigalens/src:claude/DESI238/model NCCL_NVLS_ENABLE=0
GIGALENS_DEVICE=0 python claude/DESI238/model/run_fit.py \
    --label bulgedisk_long --seed 0 --lens-light-n-max 10 \
    --n-source 2 --n-lens-light 2 --max-leapfrog 60 \
    --num-results 3000 --burnin 1000
python claude/DESI238/model/diagnose.py    claude/DESI238/model/runs/bulgedisk_long
python claude/DESI238/model/dump_params.py claude/DESI238/model/runs/bulgedisk_long
python claude/DESI238/model/plot_corner.py claude/DESI238/model/runs/bulgedisk_long
python claude/DESI238/model/plot_source.py claude/DESI238/model/runs/bulgedisk_long
python claude/DESI238/model/plot_overview.py claude/DESI238/model/runs/bulgedisk_long
```

The fit takes $\sim$20 minutes of HMC on one H100 and holds 62.7 of 80 GB; memory, not compute, is what bounds a further increase in chain length. Full narrative notes are in `RESULTS_v3.md`, alongside the model code. The model comparison of §5 came from a scratch script that is not checked in, but its numbers are reproduced there.